

# Cutinases for Biocatalytic Polycarbonate Hydrolysis: Empowering Circular Economy

Kristina Schell, Simone Göbbels, Linda Pastor, Sergii Donets, Daniela Knopp, Oskar Bamberg, Gernot Jäger, Ulrich Schwaneberg, and Lukas Reisky\*

While synthetic polymers are indispensable for the sustainable transformation of society, the increasing production needs to be combined with adequate recycling strategies. As conventional recycling processes often result in quality losses or are complicated by the presence of additives and other polymers, there is a high need for the development of new technologies. This study presents a first efficient approach to enzymatically recycle polycarbonate (PC), a high-performance polymer. Through an extensive enzyme screening, 9 cutinases are found to exhibit

hydrolytic activity toward PC in aqueous buffer enabling the recovery of bisphenol A (BPA) as a valuable monomer under mild reaction conditions. Notably, the enzymes ThcCut1-ACCG and LCC-ICCG demonstrate excellent performance by achieving conversions of 20–40% under optimized reaction conditions. Impressively, full conversion of PC can be achieved by supplementing dimethyl sulfoxide (DMSO; 30% v/v). These findings represent an excellent foundation to develop sustainable PC recycling processes for the circular economy.

## 1. Introduction

Polymers play a vital role in modern society due to their versatile properties and wide range of applications that facilitate various aspects of daily life. Their lightweight, durable, and cost-effective nature makes them indispensable materials for numerous industries, ranging from packaging and construction to transportation and healthcare. Polycarbonates (PCs) are an important class of engineering polymers with excellent material properties, such as high transparency, impact resistance, and heat deflection temperature. The term “polycarbonate” usually refers to the main polymer type on the market that is based on bisphenol A (BPA), hereafter termed PC. PCs find widespread application as construction materials, for example, in automotive engineering, as housing materials, or as a substitute for glass.<sup>[1]</sup> The global PC market with a production of over 5000 kilo tons in 2022 is expected to grow further at an annual growth rate of 6.3% during

the forecast period 2023–2032.<sup>[2]</sup> However, the increasing production and use of PCs lead to a growing amount of polymer waste at the end of the product life cycle.<sup>[3]</sup>

This waste can be recycled, but in open-loop recycling, for example, in form of upcycling or downcycling, the properties of the recycled material often differ from those of the starting material necessitating its use in other product applications. Upcycling is the process of converting waste materials into new products of better quality or higher environmental value, while downcycling involves converting waste into new materials or products of lower quality. Open-loop recycling of PC was used for the production of aliphatic polycarbonates, polyimines, and monoaryls.<sup>[4–7]</sup> Closed-loop recycling allows to reuse the materials in an identical role as before recycling. Conventional closed-loop recycling processes, such as mechanical recycling are used for a small fraction of relatively pure PC waste and have challenges when additives and other polymers are present.<sup>[8]</sup> A promising alternative is depolymerization, in which polymers are cleaved into their monomer building blocks. These monomers can subsequently be purified and reused to produce virgin-like polymer, reducing the use of fossil oil as feedstock and thereby contributing to a circular economy.<sup>[9]</sup>

The chemical recycling of PC to its valuable monomer BPA was investigated over the last decades with potential recycling routes based on hydrolysis, alcoholysis, aminolysis, and hydrogenolysis.<sup>[10]</sup> While these reactions are efficient, chemical recycling is challenged by its high energy consumption, harsh reaction conditions, often employing additional reactants, and organic solvents.<sup>[10]</sup>

A promising alternative yielding defined monomeric and oligomeric products under energy-efficient process conditions in water is the enzymatic recycling. Enzymatic polymer recycling in closed loops has been successfully demonstrated for polyethylene terephthalate (PET). Currently, an academic and industrial race is on polyamide (PA) and polyurethane (PU) recycling.<sup>[11,12]</sup>

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Looking at the global production capacity, PC has a comparable relevance as PU, PET, and PA.<sup>[2]</sup> However, studies on the enzymatic depolymerization of PCs remain scarce. In the 1990s, Suyama and Tokiwa reported on the degradation of the aliphatic polycarbonate poly(tetramethylene carbonate) using a lipoprotein lipase from *Pseudomonas* sp.<sup>[13]</sup> The polymer could be completely degraded in an aqueous suspension, yielding 1,4-butanediol and di(4-hydroxybutyl)carbonate as the reaction products.<sup>[13]</sup> For the aromatic polycarbonates, especially those based on BPA, which are economically very important, technically feasible enzymatic degradation processes have not yet been described.<sup>[14]</sup> This observation aligns with findings from enzymatic polyester degradation studies, where the depolymerization of aromatic polyesters such as PET proves to be more challenging compared to the degradation of aliphatic types such as polycaprolactone.<sup>[15]</sup>

We found the first hints in the literature for the susceptibility of PC to enzymatic hydrolysis, but so far no enzyme with significant BPA formation in aqueous medium has been described.<sup>[16]</sup> In detail, PC film weight loss was 2.5% after 30 d incubation by a bacterium, *Pseudoxanthomonas* sp.<sup>[17]</sup> It has not been reported which type of enzymes is responsible for the degradation, and whether it is a single enzyme or combined action of several enzymes. It has been shown in literature that lipases, such as those from *Candida rugosa* or *Candida antarctica*, can release BPA from PC dissolved in organic solvents. In solvents such as benzene and tetrahydrofuran polymer chains are easily accessible to enzymes.<sup>[18]</sup> In another study, enzymes were computationally designed by introducing catalytic activity toward PC into a thermostable protein fold. This resulted in a variation of the

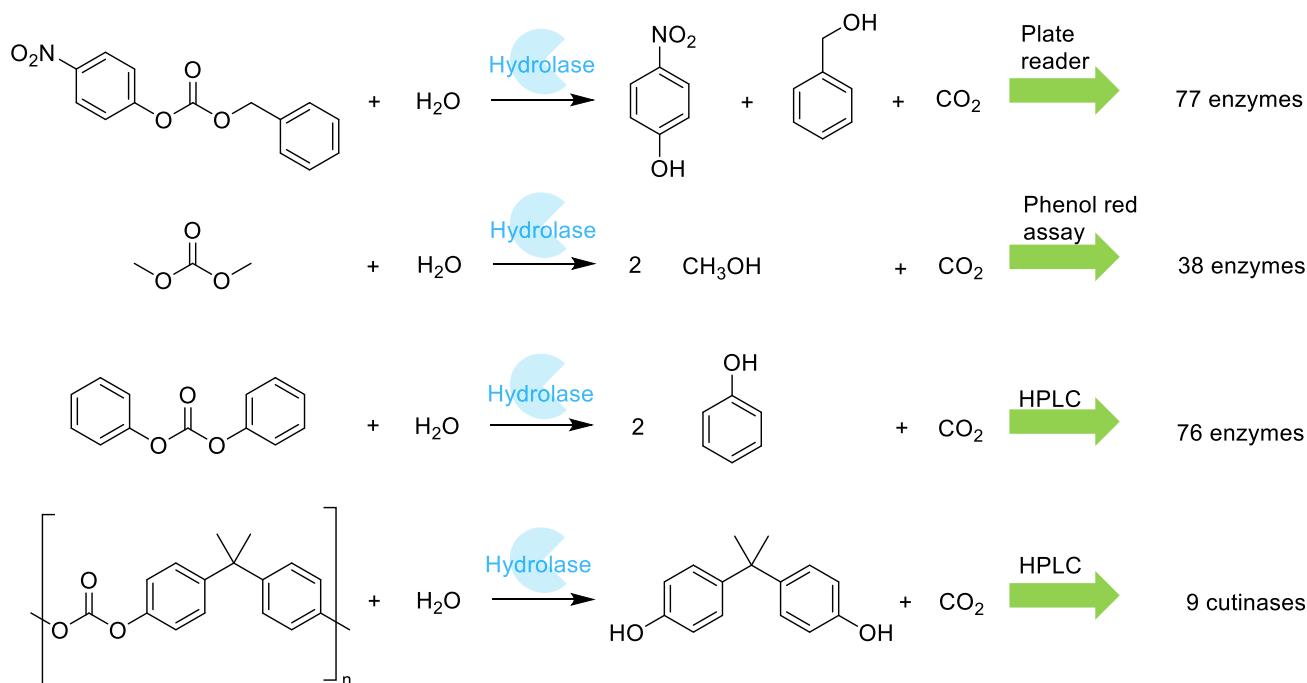
surface roughness of the PC film for the designed enzyme and implies PC degradation.<sup>[16]</sup> However, for none of the described studies on PC degradation, BPA formation was quantified. Whether hydrolytic enzymes can be employed in the significant degradation of aromatic polycarbonates was investigated in the present study.

## 2. Results and Discussion

### 2.1. Identification of 9 PC Hydrolases by Cascaded Screening

The primary aim of this study was to identify enzymes capable of specifically cleaving the carbonate bond in PC in aqueous suspension, thereby generating BPA as a monomer. The reaction conditions were optimized to facilitate efficient BPA release, and the enzymes were evaluated based on their performance in the hydrolytic cleavage of PC.

Given that there are no known naturally occurring enzymes specifically reported for PC degradation, we conducted a comprehensive screening of a diverse array of hydrolytic enzymes from EC 3 to identify potential PC-degrading biocatalysts. Our enzyme library comprised both commercial, literature based, and proprietary noncommercial enzymes from our in-house collection, encompassing a broad spectrum of hydrolases such as esterases, amidases, and urethanases. Esterases, known for their ability to hydrolyze ester bonds in both soluble substrates and solid polymers, were considered promising candidates due to the structural similarity between the ester and carbonate bond. Despite the absence of amide bonds in PC, we included amidases in our



**Figure 1.** Enzyme screening of the enzyme collection to identify carbonate hydrolases based on high (phenol red and plate reader) and low throughput (HPLC) assays with BNPC, DMC, DPC, and PC. The reactions contained ≈0.4 mg lyophilized crude lysate in 50 mM potassium phosphate buffer, 100 mM NaCl, pH 7.5.

screening based on the known promiscuity of some amidases toward ester bonds and the potential for these enzymes to exhibit unexpected activity toward carbonate linkages. Urethanasases were included as potential candidates for PC hydrolysis given their specificity for carbamate bonds and the structural similarity between the carbamate and carbonate bond, suggesting a possibility for cross-reactivity.

A cascaded screening was performed using three different carbonate substrates: the two aromatic compounds benzyl-4-nitrophenyl carbonate (BNPC) and diphenyl carbonate (DPC), and the aliphatic compound, dimethyl carbonate (DMC) (Figure 1, Figure S1, Supporting Information). As a first step, the hydrolysis of the model substrates BNPC and DMC was assayed. While the hydrolysis of BNPC was monitored by an absorbance assay, a phenol red-based assay was used to detect the pH shift resulting from the release of CO<sub>2</sub> from the carbonate hydrolysis in DMC (Figure 1). Enzymes with promising activities in initial screenings with one of the two substrates BNPC and DMC were tested with the bulky, nonfunctionalized DPC as the next step. DPC hydrolysis to phenol was measured by high-performance liquid chromatography (HPLC). Overall, 77 and 38 enzymes of the in-house enzyme collection were hydrolyzing BNPC and DMC, respectively. Seventy-six enzymes of the enzyme collection were shown to hydrolyze DPC (Figure 1, Table S1, Supporting Information). Notably, almost all enzymes hydrolyzing DMC or BNPC were able to hydrolyze DPC. The identified carbonate hydrolases were subsequently tested in a reaction with the water-insoluble polymeric PC, and BPA formation was quantified by HPLC. Remarkably, only 9 of the identified carbonate hydrolases were able to hydrolyze PC (Figure 1, Supporting Information). The systematic enzyme selection and screening aimed to maximize the likelihood of identifying novel enzymes capable of efficient PC hydrolysis.

All active enzymes were cutinases, which have been widely studied for the recycling of polyesters. Cutinases act as serine hydrolases and possess a surface-exposed active site, allowing them to act on polymer surfaces, which is in agreement with their activity on solid PC in an aqueous buffer.<sup>[19]</sup> This indicates that the access to the polymeric substrate is a major structural requirement for enzymatic PC hydrolysis and may guide future enzyme engineering efforts in this field. Serine hydrolases, including cutinases, are known for cleaving ester bonds, but no specific PC hydrolases have been identified.<sup>[20]</sup> The mechanism for carbonate hydrolysis is expected to be similar to ester hydrolysis. The reaction likely involves a nucleophilic attack by the catalytic serine on the carbonate bond, finally releasing BPA and CO<sub>2</sub>.<sup>[21]</sup> Cutinases' ability to hydrolyze carbonate bonds may be due to their open active site accommodating the bulky polycarbonate structure.

## 2.2. Optimization of Reaction Conditions Improves Conversion to BPA

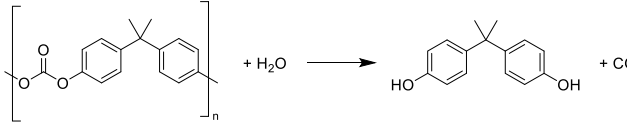
Previous studies have demonstrated that increasing the polymer surface area promotes enzymatic hydrolysis of PET.<sup>[22–24]</sup> To investigate the influence of surface area on the enzymatic hydrolysis of

PC, different particle sizes were tested. Commercial granules were compared to films and ground PC (500 μm) and three selected enzymes (Novozym®51032, ThcCut1-ACCG, and SvCut1190\*SS) were studied for their hydrolyzing activity against PC. The conversion of PC granules to the monomer BPA was <0.5% for all three enzymes. By using PC films in the reaction, 1.1–7.8% conversion was obtained. Further application of ground PC resulted in the best conversions, with up to 26.90% and 20.4% observed with Novozym®51032 and ThcCut1-ACCG, respectively (Table S2, Supporting Information). For all subsequent experiments, ground PC (500 μm) was used unless stated otherwise. The most promising enzymes hydrolyzing PC were purified, and the conversion to BPA was quantified. Minor conversions of <0.6% were observed with ThcCut1, PesH1, LCC, Novozym®51032, and BhrPETase. SvCut190\*SS, LCC-WCCG, and ThcCut1-ACCG converted 1%–5% to BPA. The highest conversion to BPA of 11.5% was observed for LCC-ICCG (Table 1). Novozym®51032 showed the highest conversion in the initial experiment without normalization of enzyme concentration but after normalizing the enzyme concentration, LCC-ICCG, LCC-WCCG, ThcCut1-ACCG, and SvCut1190\*SS exhibited higher conversion. LCC, LCC-ICCG, LCC-WCCG, and ThcCut1-ACCG were selected for further detailed characterization of the reaction parameters.

Surprisingly, ThcCut1, which differs from ThcCut1-ACCG by only 4 amino acid substitutions, showed no conversion. This indicates a mandatory effect on the conversion to BPA by these 4 mutations, which are reported to increase the T<sub>m</sub> and thermotolerance.<sup>[25,26]</sup>

LCC and LCC-ICCG tolerated reaction temperatures up to 70 °C, while LCC-WCCG exhibited decreased conversion at 70 °C with the highest conversion at 60 °C (Table 2). Notably, ThcCut1-ACCG showed minor conversions at 60 °C. One general

**Table 1.** PC hydrolysis to BPA and CO<sub>2</sub> by cutinases identified in the screening for carbonate hydrolases. Reactions contained 1% w/v PC, 100 μg ml<sup>−1</sup> purified enzyme, or 10 mg ml<sup>−1</sup> lyophilized crude lysate in 100 mM KPi buffer pH 7 and were carried out at 70 °C for 72 h in triplicates ± standard deviation.

|  |                             |
|--------------------------------------------------------------------------------------|-----------------------------|
| Enzyme                                                                               | Conversion [%]              |
| ThcCut1 <sup>[25]</sup>                                                              | 0                           |
| ThcCut1-ACCG <sup>[26]</sup>                                                         | 4.75 ± 0.01                 |
| PesH1 <sup>[36]</sup>                                                                | 0.22 ± 0.22                 |
| SvCut190*SS <sup>[37]</sup>                                                          | 1.34 ± 0.45 <sup>a,b)</sup> |
| LCC <sup>[38]</sup>                                                                  | 0.45 ± 0.01                 |
| LCC-ICCG <sup>[39]</sup>                                                             | 11.51 ± 7.29                |
| LCC-WCCG <sup>[39]</sup>                                                             | 2.90 ± 3.68                 |
| BhrPETase <sup>[40]</sup>                                                            | 0 <sup>b)</sup>             |
| Novozym®51032 <sup>[27,41]</sup>                                                     | 0.52 ± 0.13                 |

<sup>a)</sup>Determined at 60 °C. <sup>b)</sup>Determined with lyophilized lysate.

**Table 2.** Temperature and pH with the highest conversion to BPA determined with 1% w/v PC and 100  $\mu\text{g mL}^{-1}$  purified enzyme after 72 h incubation.

| Enzyme       | Opt. temperature [ $^{\circ}\text{C}$ ] | Opt. pH |
|--------------|-----------------------------------------|---------|
| ThcCut1-ACCG | 70                                      | 7       |
| LCC          | 60                                      | 9       |
| LCC-ICCG     | 70                                      | 9       |
| LCC-WCCG     | 60                                      | 9       |

trend observed was that conversions above 30% were achieved only at temperatures of 70  $^{\circ}\text{C}$  and 80  $^{\circ}\text{C}$ . At 90  $^{\circ}\text{C}$ , conversion was <5% for all four enzymes, due to thermal inactivation (Figure S2, Supporting Information). This was also observed for PET hydrolysis with cutinases.<sup>[27–29]</sup> Observed increases in conversion at higher temperatures align with the Arrhenius principle, but the sharp decline at 90  $^{\circ}\text{C}$  highlights the delicate balance between enhanced reaction kinetics and thermal tolerance of the enzyme. Notably, the conversions were in general higher than in the previous screening where only crude lysate was applied instead of purified enzyme.

The pH optimum was investigated in the range of pH 4–10 using KPi buffer, Bis-Tris propane buffer, and CHES. No activity at pH 4 could be determined for all four enzymes (Figure S3, Supporting Information). LCC and LCC-ICCG showed a similar pH profile, with constant conversions around 15%–20% in a pH range from 5 to 8. The highest conversion was determined at pH 9. LCC-ICCG has the broadest pH profile from pH 5–10, while the other tested enzymes tolerate 100 mM buffer at pH 9.5 at most. LCC-WCCG showed the highest activity at pH 5 and 9 with 17% conversion. With ThcCut1-ACCG, >30% conversion was determined at pH 7 and pH 9. Subsequent experiments were performed at the optimal pH and temperature conditions for each enzyme (Table 2). Notably, the optimal pH for ThcCut1-ACCG for BNPC hydrolysis was pH 8 (Figure S4, Supporting Information). This indicates differences in the reaction optimum associated with the enzyme's interaction with the PC substrate.<sup>[27,29]</sup> Overall, optimization of reaction conditions resulted in up to 6.8-fold increase in conversion for LCC-ICCG and 4.9-fold increase for ThcCut1-ACCG (Table S3, Supporting Information).

To correlate the temperature and pH profiles of the enzymes with their structural stability, the  $T_m$  was determined (Table S4, Figure S5, Supporting Information). Conditions with high conversions to BPA correlate with high thermal stability across various pH conditions observed for ThcCut1-ACCG and LCC-ICCG.

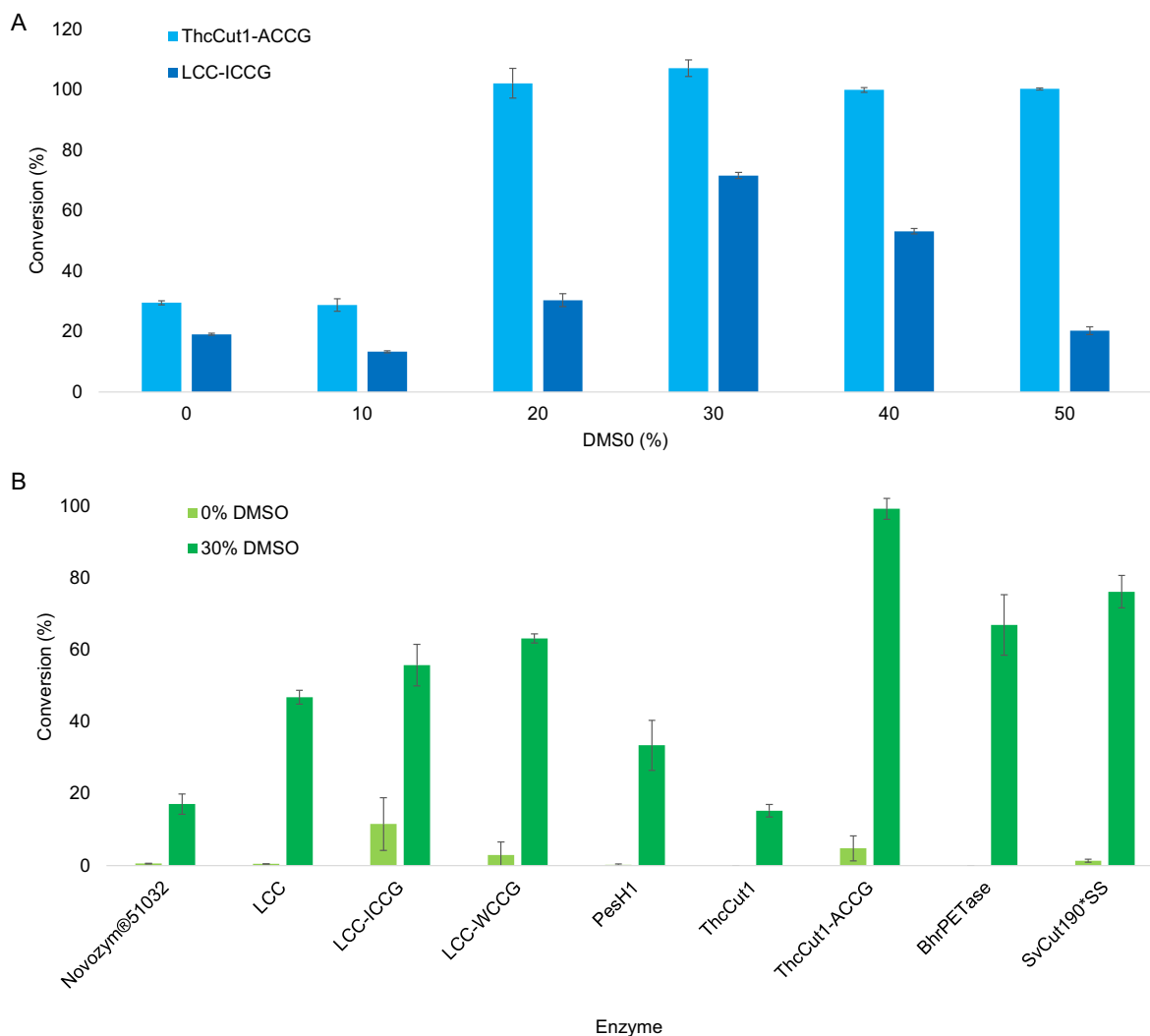
After identifying the optimal reaction conditions, it was investigated why the conversion stops at 30%–40%. Product inhibition by the formed BPA could explain the observed limitations. To test our hypothesis in reactions closer to industrial relevant conditions, BPA was supplemented to the reactions at a concentration of 7.3 mg  $\text{mL}^{-1}$ . Following a 72-hour incubation period at 70  $^{\circ}\text{C}$ , the BPA concentration in LCC, LCC-ICCG, and LCC-WCCG samples remained approximately equal to the initially supplemented amount, indicating inhibition of PC hydrolysis for these enzymes

(Figure S6, Supporting Information). In contrast, ThcCut1-ACCG samples did not exhibit such an inhibitory effect. For ThcCut1-ACCG, the final BPA concentration was approximately equal to the sum of the supplemented BPA and the BPA observed in the hydrolysis reaction without supplementation. When accounting for the supplemented BPA, the conversion in the reaction with BPA present was 25%, compared to 29.8% without BPA supplementation (Figure S6, Supporting Information). These results suggest that ThcCut1-ACCG may allow for continued BPA degradation even at high BPA concentrations, potentially offering an advantage over the other tested enzymes in environments at elevated BPA levels.

### 2.3. Addition of DMSO Results in Full Conversion

It is known that organic solvents can improve the accessibility of polymer chains for enzymes.<sup>[30]</sup> Therefore, it was tested if the addition of DMSO, toluene, or 1-butanol could improve the conversion to BPA. Addition of 10% and 20% v/v of toluene and 1-butanol did not improve the cutinase performances (Figure S7, Supporting Information). However, conversions with LCC-ICCG and ThcCut1-ACCG were increased by addition of DMSO. Full conversion to BPA with ThcCut1-ACCG could be obtained by the addition of 20%–50% v/v DMSO (Figure 2A). These results indicate that ThcCut1-ACCG demonstrates superior catalytic efficacy for hydrolysis to BPA among the enzymes evaluated in this study. Notably, PC did not appear to be significantly solubilized in the aqueous systems at these DMSO concentrations. With LCC-ICCG, the highest conversion of 72% was achieved when 30% v/v DMSO was added. With 10% v/v DMSO addition, similar conversions as observed without DMSO were determined for both enzymes. The experimental results suggest that DMSO has facilitated the accessibility of the enzyme to the polymeric substrate or influenced the solubility of the substrate or product. DMSO is known as a powerful organic solvent dissolving a wide variety of organic substances such as carbohydrates, synthetic polymers, and peptides.<sup>[31,32]</sup> In addition, DMSO has been reported to improve enzyme activity and stability in various enzymatic reactions, e.g., hydrolysis with lipases.<sup>[33,34]</sup> With the addition of DMSO, a remarkable 100% conversion was attained, underscoring the synergistic effects of optimized reaction conditions in facilitating efficient enzymatic hydrolysis. This achievement holds significant economic benefits for the circular economy, as maximizing conversion and reaction rates promotes resource efficiency, aligning with the principles of circularity.

With these optimized conditions, different forms of PC were compared using 0% and 30% DMSO as a cosolvent. Film and shredded PC yielded the highest overall conversions to BPA among the substrates tested (Figure S8, Supporting Information). The enzymes ThcCut1-ACCG and LCC-ICCG demonstrated superior performances with these materials, achieving conversions of >93% and >85%, respectively, in the presence of 30% DMSO. In the absence of DMSO, ThcCut1-ACCG and LCC-ICCG still exhibited the highest conversions, yielding 28% and 26% conversion, respectively. The conversions for reactions



**Figure 2.** Effect of DMSO concentration on PC hydrolysis A) by purified ThcCut1-ACCG and LCC-ICCG, and B) by enzyme lysate of 9 PC hydrolases identified in this study. Conversion to BPA by  $50 \mu\text{g ml}^{-1}$  purified ThcCut1-ACCG and LCC-ICCG in the presence of 0%–50% v/v DMSO was determined in duplicates. Enzyme lysate ( $10 \text{ mg ml}^{-1}$ ) was tested with 0% v/v and 30% v/v DMSO.

using film or shredded PC were 1.3- to 6.8-fold higher compared to those using PC granules. Under these optimized conditions, no significant differences were observed between film and shredded material, as both achieved complete conversion at the endpoint measurement.

Furthermore, it was verified that the BPA does not exhibit a significant inhibitory effect on ThcCut1-ACCG under these conditions (Figure S9, Supporting Information).

Subsequently, we sought to elucidate the factors contributing to the enhanced conversion to BPA in the presence of DMSO. The observed positive effect of DMSO could be attributed to an acceleration of the catalytic carbonate cleavage by the enzyme or an improved accessibility of the carbonate bonds within the insoluble polymer.

To assess this hypothesis, the specific activity was determined for the soluble carbonate substrate BNPC and for PC with and without DMSO (Table 3). While DMSO increased the specific activity with PC 48-fold, the specific activity with BNPC increased only

**Table 3.** Specific activity of ThcCut1-ACCG with BNPC and PC supplemented with 0% and 30% DMSO. NP, 4-nitrophenol.

|          | BNPC                                                                                       | PC                                                                                          |
|----------|--------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| DMSO [%] | Spec. activity<br>[ $\text{mg}_{\text{NP}} \text{h}^{-1} \text{mg}_{\text{Enzyme}}^{-1}$ ] | Spec. activity<br>[ $\text{mg}_{\text{BPA}} \text{h}^{-1} \text{mg}_{\text{Enzyme}}^{-1}$ ] |
| 0        | 1846.3                                                                                     | 5.4                                                                                         |
| 30       | 3018.3                                                                                     | 86                                                                                          |

1.6-fold. This suggests that the positive effect of DMSO on the hydrolysis of polycarbonate does not primarily result from an improved capability to hydrolyze the carbonate bond but from an improved accessibility of these bonds within the polymer.

To further investigate this hypothesis, molecular dynamics simulations were conducted with PC in environments with varied solvent concentrations. The simulations revealed that DMSO can penetrate into the PC after  $2 \mu\text{s}$  of simulation time (Figure S10A, Supporting Information). This indicates a weakening of direct

interactions between polymer chains (Figure S10A). Further comparative simulations were performed in water and in a water-DMSO mixture (70:30 v/v). Results showed that PC maintains a more compact conformation in pure water compared to the water-DMSO mixture (Figure S10B, Supporting Information). This suggests that the presence of DMSO promotes an expanded PC structure, promoting accessibility for an enzymatic attack.<sup>[35]</sup> To explore enzyme-substrate interactions, ThcCut1-ACCG was included into the simulations. The distance between the enzyme's catalytic serine and the carbonate bonds of PC under different solvent conditions was calculated. The simulations demonstrated that DMSO in the reaction allows for the enzyme to obtain close proximity to the PC. Importantly, the DMSO solvent effect resulted in a reduced distance between the carbonate bonds of PC and the enzyme's catalytic serine upon binding of the enzyme to the surface (Figure S10C, Supporting Information). This decreased spatial separation likely facilitates the hydrolysis reaction by bringing the reactive groups into closer proximity. The simulations suggest that DMSO plays a crucial role in modulating PC's structure and its interactions with the hydrolytic enzyme, thereby promoting more efficient catalysis.

To validate the general principle of DMSO boosting PC degradability, the lysates of the 9 enzymes that initially exhibited hydrolytic activity toward PC were tested without DMSO or with 30% v/v DMSO. For all enzymes, the addition of DMSO led to a higher conversion from 15% to 99% (Figure 2B). The best enzyme in this test was ThcCut1-ACCG with 99% conversion upon the addition of 30% v/v DMSO.

In conclusion, the presence of DMSO remarkably facilitated the enzymatic depolymerization of PC, enabling complete conversion to monomers. Furthermore, it substantially enhanced the specific activity of ThcCut1-ACCG toward PC, resulting in a 48-fold increase compared to reactions without DMSO. These findings highlight the potential of DMSO as a valuable additive to improve the efficiency of enzymatic recycling processes for PC waste.

While the current study demonstrates high efficiency of ThcCut1-ACCG under optimized conditions, further enzyme engineering could potentially be employed to enhance enzyme stability and/or activity under process conditions.

### 3. Conclusion

In summary, this study identified highly efficient PC-hydrolyzing enzymes and describes a boosting effect of DMSO to quantitatively convert PC into its monomer BPA under mild conditions (70 °C, pH 7). The cutinase variants ThcCut1-ACCG and LCC-ICCG demonstrated particularly high activity, achieving up to 100% conversion to BPA under optimal conditions. These findings pave the way for further development of enzymatic processes for sustainable PC recycling and BPA recovery, contributing to the transition toward a circular economy and addressing the growing environmental challenges associated with polymer waste.

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### Conflict of Interest

The authors declare no conflicts of interest.

### Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

**Keywords:** cutinases • enzymatic recycling • hydrolysis • polycarbonates • polymers

- [1] J. L. DeRudder, *Handbook of Polycarbonate Science and Technology*, 1st ed. (Ed.: J. T. Bendler), CRC Press, Boca Raton **2000**, 303.
- [2] Prismane Consulting, GMB - 021 **2023**, 1.
- [3] B. J. Seewoo, E. V. S. Wong, Y. R. Mulders, L. M. Goodes, E. Eroglu, M. Brunner, A. Gozt, P. Toshniwal, C. Symeonides, S. A. Dunlop, *Heliyon* **2024**, *10*, e32912.
- [4] K. M. Derr, R. C. Smith, *J. Polym. Sci.* **2023**, *62*, 1115.
- [5] N. Kasmi, E. Bäckström, M. Hakkarainen, *Resour. Conserv. Recycl.* **2023**, *193*, 106974.
- [6] K. S. K. Reddy, Y.-C. Chen, R.-J. Jeng, M. M. Abu-Omar, C.-H. Lin, *ACS Sustain. Chem. Eng.* **2023**, *11*, 8580.
- [7] K. Saito, C. Jehanno, L. Meabe, J. L. Olmedo-Martínez, D. Mecerreyes, K. Fukushima, H. Sardon, *J. Mater. Chem. A* **2020**, *8*, 13921.
- [8] M. Auer, J. Schmidt, J. Diemert, G. Gerhardt, M. Renz, V. Galler, J. Woidasky, *Recycling* **2023**, *8*, 18.
- [9] S. Thiagarajan, E. Maaskant-Reilink, T. A. Ewing, M. K. Julsing, J. van Haveren, *RSC Adv.* **2021**, *12*, 947.
- [10] J. G. Kim, *Polym. Chem.* **2020**, *11*, 4830.
- [11] a) Y. Branson, S. Sörtl, C. Buchmann, R. Wei, L. Schaffert, C. P. S. Badenhorst, L. Reisky, G. Jäger, U. T. Bornscheuer, *Angew. Chem. Int. Ed.* **2023**, *62*, e202216220; b) Y. Branson, S. Sörtl, C. Buchmann, R. Wei, L. Schaffert, C. P. S. Badenhorst, L. Reisky, G. Jäger, U. T. Bornscheuer, *Angew. Chem.* **2023**, *135*, e202216220.
- [12] E. L. Bell, G. Rosetto, M. A. Ingraham, K. J. Ramirez, C. Lincoln, R. W. Clarke, J. E. Gado, J. L. Lilly, K. H. Kucharzyk, E. Erickson, G. T. Beckham, *Nat. Commun.* **2024**, *15*, 1217.
- [13] T. Suyama, Y. Tokiwa, *Enzyme Microb. Technol.* **1997**, *20*, 122.
- [14] A. Kausar, *J. Plast. Film Sheeting* **2018**, *34*, 60.
- [15] C. Gamerith, M. Vastano, S. M. Ghorbanpour, S. Zitzenbacher, D. Ribitsch, M. T. Zumstein, M. Sander, E. Herrero Acero, A. Pellis, G. M. Guebitz, *Front. Microbiol.* **2017**, *8*, 938.
- [16] L. H. Holst, N. G. Madsen, F. T. Toftgård, F. Rønne, I. M. Moise, E. I. Petersen, P. Fojan, *Protein Eng. Des. Sel.* **2023**, *36*, gzad022.
- [17] W. Yue, C.-F. Yin, L. Sun, J. Zhang, Y. Xu, N.-Y. Zhou, *J. Hazard. Mater.* **2021**, *416*, 125775.
- [18] G. Sivalingam, G. Madras, *J. Appl. Polym. Sci.* **2003**, *91*, 2391.
- [19] V. Tournier, S. Duquesne, F. Guillamot, H. Cramail, D. Taton, A. Marty, I. André, *Chem. Rev.* **2023**, *123*, 5612.
- [20] M. Holmquist, *Curr. Protein Peptide Sci.* **2000**, *1*, 209.
- [21] R.-T. Guo, X. Li, Y. Yang, J.-W. Huang, P. Shen, R. K. Liew, C.-C. Chen, *Environ. Chem. Lett.* **2024**, *22*, 1275.
- [22] F. Awaja, D. Pavel, *Eur. Polym. J.* **2005**, *41*, 1453.
- [23] E. S. Barboza Neto, L. A. F. Coelho, M. M. C. Forte, S. C. Amico, C. A. Ferreira, *Mater. Res.* **2013**, *17*, 236.

- [24] B. Ciuffi, E. Fratini, L. Rosi, *Polym. Degrad. Stab.* **2024**, 222, 110698.
- [25] D. Ribitsch, A. Hromic, S. Zitzenbacher, B. Zartl, C. Gamerith, A. Pellis, A. Jungbauer, A. Łyskowski, G. Steinkellner, K. Gruber, R. Tscheliessnig, E. Herrero Acero, G. M. Guebitz, *Biotechnol. Bioeng.* **2017**, 114, 2481.
- [26] Z. Zhang, S. Huang, D. Cai, C. Shao, C. Zhang, J. Zhou, Z. Cui, T. He, C. Chen, B. Chen, T. Tan, *Green Chem.* **2022**, 24, 5998.
- [27] M. Ronkvist, W. Xie, W. Lu, R. A. Gross, *Macromolecules* **2009**, 42, 5128.
- [28] A. N. Shirke, C. White, J. A. Englaender, A. Zwarycz, G. L. Butterfoss, R. J. Linhardt, R. A. Gross, *Biochemistry* **2018**, 57, 1190.
- [29] E. Erickson, J. E. Gado, L. Avilan, F. Bratti, R. K. Brizendine, P. A. Cox, R. Gill, R. Graham, D. J. Kim, G. Konig, W. E. Michener, S. Poudel, K. J. Ramirez, T. J. Shakespeare, M. Zahn, E. S. Boyd, C. M. Payne, J. L. DuBois, A. R. Pickford, G. T. Beckham, J. E. McGeehan, *Nat Commun* **2022**, 13, 7850.
- [30] C. Silva, T. Matamá, G. M. Guebitz, A. Cavaco-Paulo, *J. Polym. Sci. A Polym. Chem.* **2005**, 43, 2749.
- [31] Y. Eom, B. C. Kim, *Polymer* **2014**, 55, 2570.
- [32] M. Sun, Z. Xu, N. E. Munyaneza, Y. Zhang, C. Posada, G. Liu, *Polym. Chem.* **2023**, 14, 1915.
- [33] D. S.-H. Chan, M. E. Kavanagh, K. J. McLean, A. W. Munro, D. Matak-Vinkovic, A. G. Coyne, C. Abell, *Anal. Chem.* **2017**, 89, 9976.
- [34] A. N. Rajeshwara, V. Prakash, *Int. J. Pept. Protein Res.* **1994**, 44, 435.
- [35] Q. Wang, J. Cai, L. Zhang, M. Xu, H. Cheng, C. C. Han, S. Kuga, J. Xiao, R. Xiao, *J. Mater. Chem. A* **2013**, 1.
- [36] L. Pfaff, J. Gao, Z. Li, A. Jackering, G. Weber, J. Mican, Y. Chen, W. Dong, X. Han, C. G. Feiler, Y.-F. Ao, C. P. S. Badenhorst, D. Bednar, G. J. Palm, M. Lammers, J. Damborsky, B. Strodel, W. Liu, U. T. Bornscheuer, R. Wei, *ACS Catal.* **2022**, 12, 9790.
- [37] M. Oda, Y. Yamagami, S. Inaba, T. Oida, M. Yamamoto, S. Kitajima, F. Kawai, *Appl. Microbiol. Biotechnol.* **2018**, 102, 10067.
- [38] S. Sulaiman, S. Yamato, E. Kanaya, J.-J. Kim, Y. Koga, K. Takano, S. Kanaya, *Appl. Environ. Microbiol.* **2012**, 78, 1556.
- [39] V. Tournier, C. M. Topham, A. Gilles, B. David, C. Folgoas, E. Moya-Leclair, E. Kamionka, M.-L. Desrousseaux, H. Texier, S. Gavalda, M. Cot, E. Guémard, M. Dalibey, J. Nomme, G. Cioci, S. Barbe, M. Chateau, I. André, S. Duquesne, A. Marty, *Nature* **2020**, 580, 216.
- [40] X. Xi, K. Ni, H. Hao, Y. Shang, B. Zhao, Z. Qian, *Enzyme Microb. Technol.* **2021**, 143, 109715.
- [41] R. Brackmann, C. de Oliveira Veloso, A. M. de Castro, M. A. P. Langone, *3 Biotech* **2023**, 13, 135.

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